

Label-Free Health Ranking of Power Transformers from Dissolved Gas Analysis

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End-of-project report — Weeks 1 to 5 — 17 July 2026

Executive summary. Power transformers fail expensively, and Dissolved Gas Analysis (DGA) is the standard way to catch faults early. Conventional interpretation (Duval, IEC, Rogers) is rule-based, needs an expert, and does not scale to a whole fleet without labels. This project developed a **label-free, self-supervised** framework that **ranks a fleet of transformers by incipient-fault risk** from their DGA history, and explains each rank by separating fault *type* from *severity*. On a real longitudinal fleet (628 units, 4,563 samples, ~4.5 years), the label-free risk ranking reaches **AUC 0.74** against recorded field events (vs 0.52 for a total-gas baseline), and the compositional representation **triples** the agreement with expert fault types (ARI 0.14 to 0.47).

1. Context and motivation

Power transformers are among the most critical and expensive assets of an electrical grid; an in-service failure means outages, safety risk and costly, slow replacement. **Dissolved Gas Analysis (DGA)** detects incipient faults early: internal thermal or electrical stress decomposes the oil and paper insulation and releases characteristic gases (hydrogen, methane, ethylene, acetylene, carbon oxides) measured in ppm. Conventional interpretation methods — IEC 60599 ratios, Rogers ratios and the Duval Triangle — are **rule-based**, require **expert judgement**, sometimes return “**not determined**”, and implicitly assume a per-sample fault *label* that utilities rarely have at scale. In practice an operator must **prioritise maintenance across an entire fleet** — rank units by risk — **without labels**. That is the gap this project addresses.

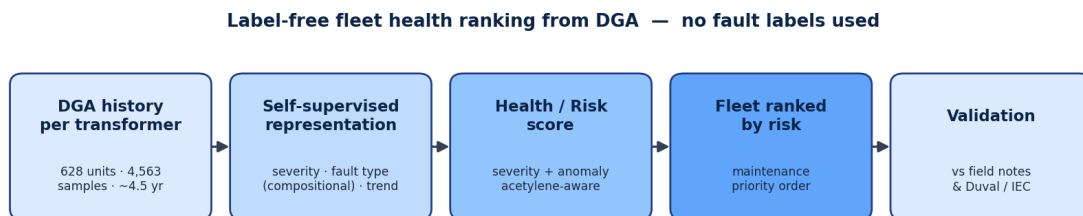
2. Objective, research question and dataset

Research question. Can we rank transformers across a fleet by incipient-fault risk, *label-free*, by learning a self-supervised, temporal representation of each unit’s DGA history and scoring severity and deviation (acetylene-aware) — and how does this compare with conventional DGA interpretation?

Dataset. 628 transformers, **4,563 main-tank oil samples**, 42 manufacturers, 115–525 kV, spanning a median ~4.5 years at ~6-month sampling intervals (a genuinely **longitudinal** fleet). **391 samples carry a real field-event note** (Buchholz trip, bushing failure, high power factor...), used as **weak validation labels** — sparse and imperfect, so results are read as indicative, not absolute.

3. Approach

The pipeline turns each unit's raw DGA history into a single risk rank, with no fault labels used at any stage, and is benchmarked against the conventional methods.



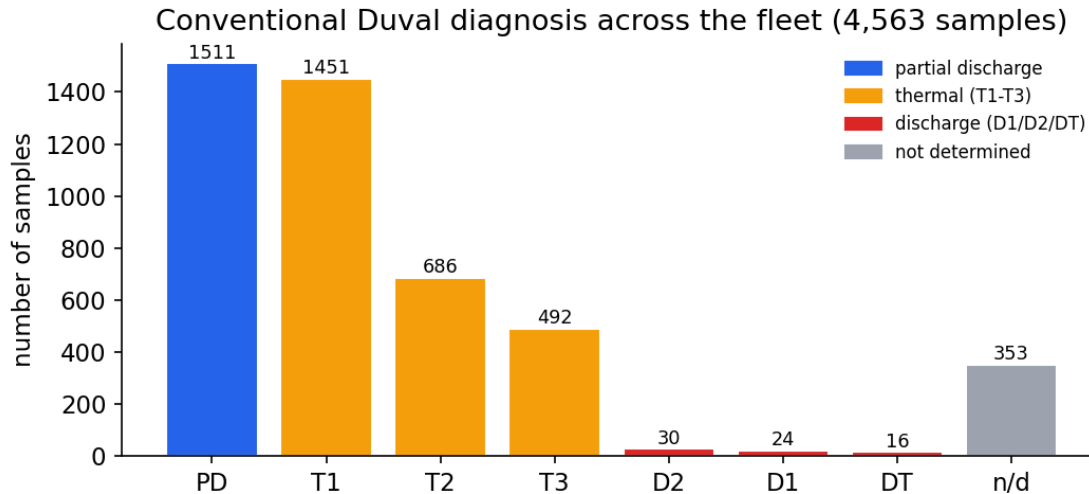
Self-supervised + temporal representation learning, validated against conventional methods (IEEE C57.104 / IEC 60599 / Rogers / Duval)

Pipeline: from each transformer's DGA history to a risk-ranked fleet, with no fault labels.

4. Week-by-week progress

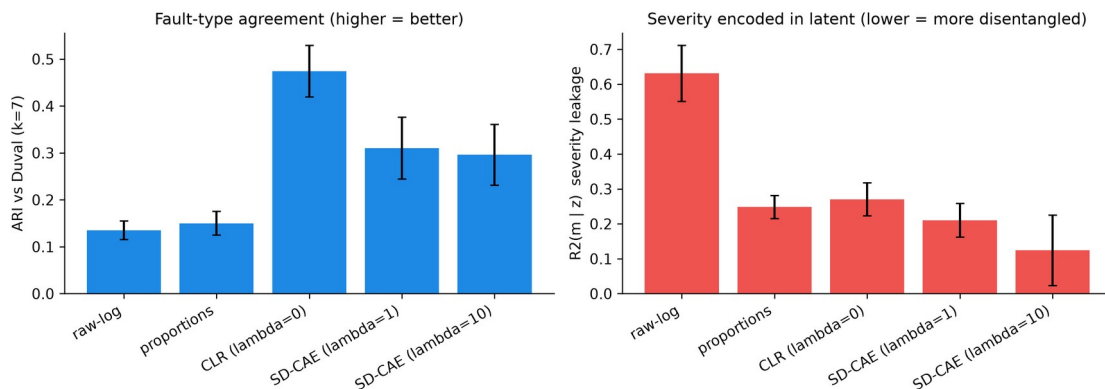
Week	Dates	Theme	Main outcome
1	15–19 Jun	Foundations	Clean data + EDA, conventional baselines, literature (16 refs + notes), problem statement
2	22–26 Jun	Representation learning	AE + VAE; compositional representation separating fault type from severity (ARI 0.14 to 0.47)
3	29 Jun–3 Jul	Clustering + anomaly detection	Fault-family clusters + anomaly scoring, validated against field-event notes
4	6–10 Jul	Health/risk index + evaluation	Label-free fleet ranking, AUC 0.74; systematic comparison vs conventional methods
5	13–17 Jul	Writing + dissemination	IEEE paper draft, frozen figures/tables, reproducible repository, defense slides

Week 1 — Foundations. Built and verified the full pipeline end-to-end on the real data: cleaning (gas values stored as text, missing markers, field notes), exploratory analysis, and the conventional baselines (Duval / IEC / Rogers) reproduced as the reference to compare against. The literature review produced 16 references with per-paper reading notes. The fleet is dominated by partial-discharge and thermal diagnoses, with a strong class imbalance to keep in mind for evaluation.



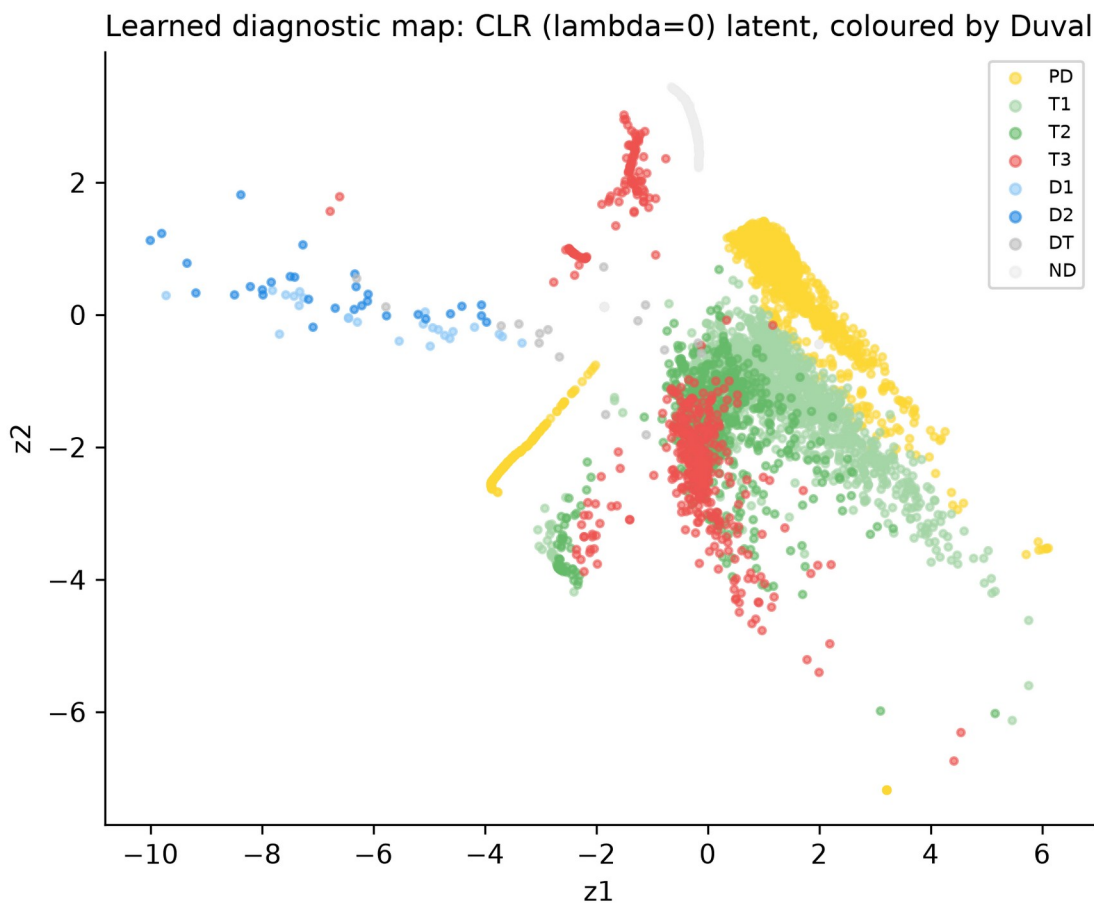
Conventional Duval diagnosis across the fleet — the baseline to compare against, with clear class imbalance.

Week 2 — Representation learning. Trained and tuned an autoencoder and compared a variational autoencoder. The key experiment separated **magnitude** (total gas = severity) from **composition** (gas proportions = fault type). Encoding only the composition (a CLR transform) raised agreement with Duval fault types from **ARI 0.14 to 0.47** (about $\times 3.4$, over 5 seeds) and removed severity leakage (R^2 0.63 to 0.27). An adversarial variant did **not** help — a negative result reported honestly.



Left: agreement with Duval fault type jumps with the compositional representation. Right: the plain model mostly encodes severity.

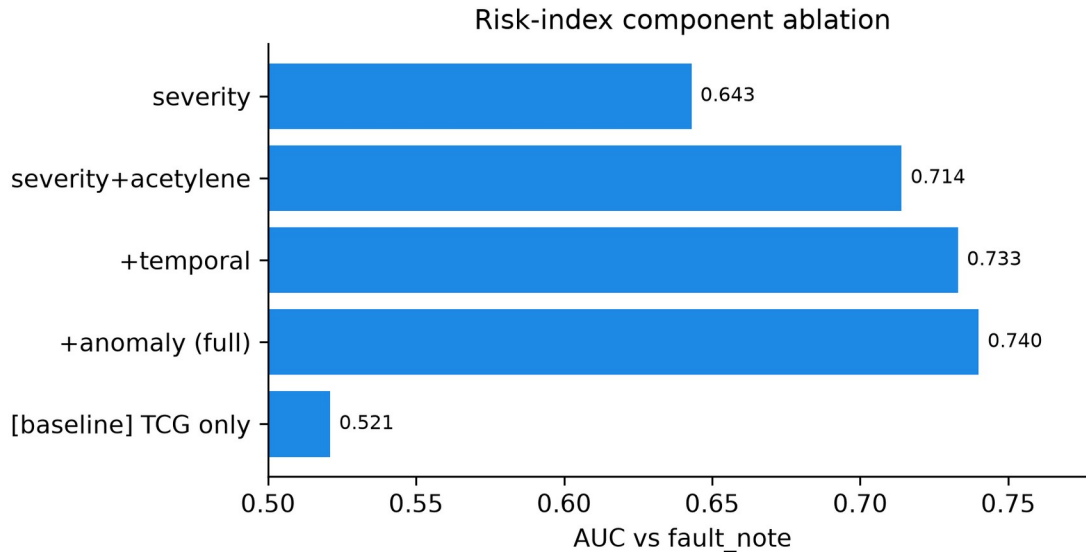
The chosen representation shows real fault-type structure — discharge, thermal and partial-discharge samples fall in different regions:



The compositional representation, coloured by Duval type: fault families separate into different regions.

Week 3 — Clustering and anomaly detection. Clustered the representation (KMeans / GMM / HDBSCAN, model selection by silhouette) and interpreted each cluster physically against the Duval mix. Added anomaly scoring by autoencoder reconstruction error and Isolation Forest, and validated flagged anomalies against the field-event notes (precision/recall reported as indicative given weak labels).

Week 4 — Health/risk index and evaluation. Combined severity, an **acetylene weighting** (acetylene = arcing = the most dangerous fault), a **temporal** gas-growth-rate feature, and the anomaly score into a single **label-free Health/Risk Index**, then ranked the fleet. Each component adds measurable value, and the full index reaches **AUC 0.74** against field events, far above a total-gas baseline (0.52).

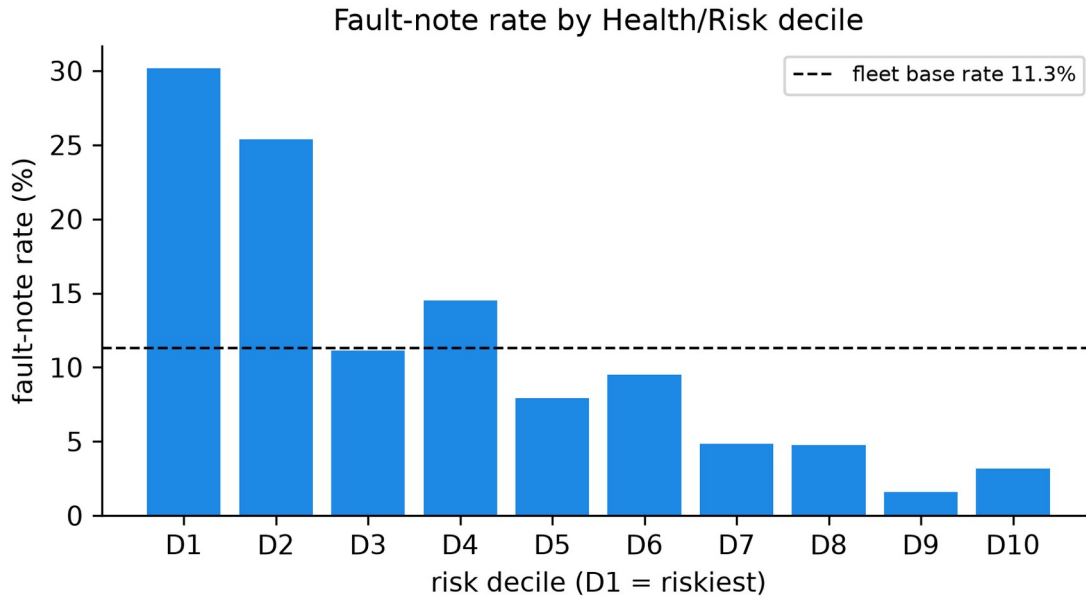


Each component of the risk index adds value: severity 0.64, then +acetylene 0.71, +temporal 0.73, +anomaly 0.74 (vs 0.52 for a total-gas baseline).

Week 5 — Writing and dissemination. Drafted the IEEE paper (all sections), froze the figures and tables under results/, finalised the reproducible repository, and prepared the defense slides.

5. Key results

The ranking is **meaningful**: the 10% riskiest units carry far more real field faults (~30%) than the fleet average (~11%) and than the safest units (~2–3%), with a monotonic trend across deciles.



Riskiest units (left, D1) carry many more real field faults than safe units (right, D10) — the ranking is meaningful.

- **Representation:** fault-type agreement tripled (ARI 0.14 to 0.47); severity leakage removed.
- **Risk ranking:** AUC 0.74 vs 0.52 baseline; monotonic risk deciles; top-decile lift up to $\times 2.7$.
- **Component value:** severity 0.64 to +acetylene 0.71 to +temporal 0.73 to +anomaly 0.74.
- **Vs conventional:** the label-free ranking recovers the units the conventional severity ordering flags, and surfaces some the rules miss — while needing no labels and no expert thresholds.

6. Conclusion

The project delivered a **label-free, self-supervised framework** that ranks a real fleet of transformers by incipient-fault risk, and explains each rank by separating fault *type* (a compositional representation) from *severity* (an acetylene-aware risk score). Three contributions stand out: (1) the label-free fleet health-ranking pipeline, validated on a longitudinal fleet; (2) a compositional representation that recovers fault type and provides the interpretable “why” behind a rank; (3) a systematic comparison with the conventional IEC / Duval / Rogers methods. Because no ground-truth health label exists, the ranking is validated **indirectly** (field-event notes + agreement with C57.104 severity), and the framework is positioned as **decision support**, not a claim to beat supervised classifiers (a different setting). In short, it turns DGA from a per-sample rule lookup into a **fleet-level, label-free maintenance-prioritisation tool**.

7. Limitations and next steps

- **Weak, sparse labels** (391 field-event notes) → validation is indicative; we report note-rate by decile and precision@k rather than a single accuracy number.
 - **Temporal features are engineered** (gas-growth rates); a deep sequence model over each unit's trajectory is the natural next step.
 - **Confound control:** the index deliberately uses physical-gas features only (not sampling frequency, which correlates with faults only because faulted units get re-sampled).
 - **Highest-leverage improvement:** a better label — translating the Thai field notes or obtaining the utility's failure records — would strengthen every risk claim.
 - **Forward direction** (consistent with the 2026 review literature): deep temporal models, Deep SVDD anomaly detection, digital-twin remaining-useful-life prognostics, and privacy-preserving federated learning across utilities.
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Reproducible: every figure and number is produced by code under results/. This is a research initiation (5 weeks, one student); results are early and honestly hedged.